Lesson 13: Experiment Basics

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 5.1: Prelude to Experimental Research; 5.2: Experiment Basics.

In this lesson

You will learn what makes a study an experiment: the researcher manipulates an independent variable and controls extraneous variables, so that only the independent variable can explain differences between conditions. You will also practise the passive voice and the language of definitions.

Learning goals

After this lesson you should be able to:

- Identify the independent variable, its conditions and the dependent variable in an experiment.
- Distinguish manipulation from control, and explain why pre-existing groups do not make an experiment.
- Explain how extraneous variables add noise or become confounding variables.
- Compare no-treatment, placebo, wait-list and best-alternative control conditions.
- Describe a procedure in the passive voice and define research terms precisely.

Key language

Term or phrase	Meaning	Example
experiment	a study in which an independent variable is manipulated and a dependent variable is measured while extraneous variables are controlled	Darley and Latané's study was an experiment.
independent variable (IV)	the variable whose changes are thought to cause a change in another variable	the number of witnesses that participants believed to be present
dependent variable (DV)	a variable that is thought to be the effect of the independent variable	whether the participant left the room to seek help
conditions	the levels of the independent variable	one, two or five other students
extraneous variable	anything that varies in a study other than the IV and the DV	writing ability, diet, time of day
control	holding extraneous variables constant	All participants were tested in the same room.

Term or phrase	Meaning	Example
confounding variable	an extraneous variable that differs across the levels of the IV	higher average IQ in one condition
random assignment	assigning participants to conditions by a random procedure, such as flipping a coin	They randomly assigned their participants to conditions.
placebo effect	the positive effect of a placebo (a simulated treatment with no active ingredient)	a "sugar pill" that helps
placebo control condition / waitlist control condition	a control condition in which participants receive a placebo / receive the treatment after the study	an identical-looking pill without the active ingredient / They must wait.
randomized clinical trial	an experiment that tests the effectiveness of a psychological or medical treatment	a trial of a new psychotherapy
in other words	introduces a clearer restatement	In other words, whether changes in one variable cause a change in another.
over and above	more than; in addition to	improvement over and above that in the placebo group

Rules and explanations

The ideas

The bystander experiment (Section 5.1)

- **Hypothesis** (Darley & Latané, 1968): the more witnesses there are to an accident or a crime, the less likely any of them is to help. **Theory:** diffusion of responsibility, meaning each witness feels less responsible.
- Real cases such as the murder of Kitty Genovese fitted the idea but proved nothing: would the witnesses have helped had there been fewer of them?
- **Method:** each student sat alone in a small room for an intercom "discussion" in which another "student" seemed to have a seizure (the comments were prerecorded). Participants were told that the discussion involved one, two or five other students.
- **Result:** helping fell from 85% to 62% to 31% as the number of "witnesses" increased. Because this was the only difference between the groups, it must have caused the difference in helping.
- **Box "The Parable of the 38 Witnesses":** the standard Genovese story is inaccurate (Manning, Levine, & Collins, 2007), and it may have drawn attention away from the conditions in which people do respond collectively.

Two fundamental features (Section 5.2)

1. **Manipulation:** the researcher systematically varies the level of the IV. Each level is a condition, often given a short name ("traumatic condition", "neutral condition"). Darley and Latané had *one* IV with *three* levels, not three IVs: a **single factor multi-level design**. One IV with two conditions is a **single factor two-level design**.
2. **Control:** the researcher minimises variability in extraneous variables (same room, same emergency) and randomly assigns participants so the groups start out similar.

In research, you **manipulate** the IV and **control** other variables; the two words are not synonyms.

Manipulation needs active intervention. Comparing people who already keep a journal with people who do not is not an experiment: journal keepers may also be more conscientious, more introverted or less stressed, and any of these could explain a difference in health. Variables that cannot be manipulated for practical or ethical reasons (such as early illness experience) need nonexperimental approaches.

Extraneous variables: noise and confounds

- **Participant variables** (writing ability, diet, gender) and **situation or task variables** (time of day, hand or computer, weather).
- **Noise:** in the mood and memory table, every happy participant in the idealised data recalls 4 happy events and every sad participant 3. The realistic data have the same means ($M = 4$, $M = 3$), but the scores vary widely, so the effect is harder to see.
- **Holding variables constant** reduces noise (same location, identical instructions, only right-handed participants in language studies), but a very narrow sample lowers **external validity**: how far results generalise beyond the people studied.
- **Confounds:** IQ varies in almost every experiment; that is acceptable if average IQ is similar across conditions. If the positive-mood group had higher IQs, nobody could tell whether mood or IQ caused its better memory. The general solution is random assignment (Lesson 14).

Treatment and control conditions

Control condition	Participants receive	Evaluation
No-treatment	nothing	The treatment group may improve only because they expect to (placebo effect).
Placebo control	something like the treatment, without the active element	Expectations are similar, so extra improvement is caused by the treatment.
Wait-list control	the treatment later	Control participants still expect to improve (eventually).
Best available alternative	an existing treatment, e.g., exposure therapy for simple phobia	Asks "Does it work better than what is already available?"

Informed consent requires telling participants that they may receive a placebo. In the box "The Powerful Placebo", knee patients given sham surgery improved as much as those given real surgery (Moseley et al., 2002).

Language focus

1. The passive voice in method descriptions. Methods focus on what was done, not who did it, so they use *be* + past participle:

- "Each of their university student participants **was isolated** in a small room and **told** that he or she would be having a discussion ..."
- "These comments **had been prerecorded** and **were played back** to create the appearance of a real emergency."

Use the present passive for a general procedure ("participants **are randomly assigned**") and the future passive in a proposal ("Participants **will be assigned**").

2. Defining terms.

- *X is called Y:* "The different levels of the independent variable **are called** conditions."
- *X (referred to as Y):* "changes in one variable (**referred to as** an independent variable)"
- *Y is a + general class + that ...:* "A placebo **is a** simulated treatment **that** lacks any active ingredient ..."

> **Note for Arabic speakers** - Arabic needs no *be* in the present, so learners sometimes drop it from the passive. "Participants **were** assigned" is a passive; "participants assigned" means the participants did the assigning. - *Research* is uncountable: "much research", "two studies", never "researches".

Practice

1. Reading questions

1. What was Darley and Latané's hypothesis, and what process did they propose to explain it?
2. In their experiment, what were the IV, the conditions and the DV?
3. Why did real cases such as Kitty Genovese's murder not provide convincing evidence?
4. A researcher compares people who already keep a journal with people who do not. Why is this not an experiment?
5. In the table of noiseless and noisy data, what are the condition means? In the noisy data, how many sad-mood participants recalled more happy events than the happy-mood mean?
6. What is the downside of limiting participants to one very specific category?
7. When does IQ become a confounding variable, and why is that a problem?
8. Why can a treatment condition and a no-treatment control condition alone not show that a treatment works?

2. Book exercises

From the chapter exercises in Section 5.6 (printed with Lesson 16).

1. Practice: List five variables that can be manipulated by the researcher in an experiment. List five variables that cannot be manipulated by the researcher in an experiment.
2. Practice: For each of the following topics, decide whether that topic could be studied using an experimental research design and explain why or why not. a. Effect of parietal lobe damage on people's ability to do basic arithmetic. b. Effect of being clinically depressed on the number of close friendships people have. c. Effect of group training on the social skills of teenagers with Asperger's syndrome. d. Effect of paying people to take an IQ test on their performance on that test.
3. Discussion: Imagine that an experiment shows that participants who receive psychodynamic therapy for a dog phobia improve more than participants in a no-treatment control group. Explain a fundamental problem with this research design and at least two ways that it might be corrected.

3. Language exercise

Rewrite in the passive voice without the agent (1-4). Then write a definition with "A ... is a ... that ..." (5).

1. The researchers isolated each participant in a small room.
2. The researchers told some participants that the discussion involved only one other student.
3. Someone had prerecorded the comments and played them back.
4. Researchers often give the conditions short descriptive names.
5. Define *confounding variable*.

Speak with your tutor

1. Explain the difference between manipulating and controlling a variable, using Darley and Latané's experiment. Ask the tutor to correct your use of *manipulate*, *control* and *condition*.
2. In PSYC 110 Lecture 15 (Morality), Professor Bloom tells the standard Kitty Genovese story ("dozens of people watched and did nothing"). Compare it with the box "The Parable of the 38 Witnesses": how can a story inspire research and mislead at the same time?
3. In PSYC 110 Lecture 9, Professor Salovey says the "rickety bridge" study is not an experiment because there was no random assignment. Explain why, using the journal-keeping example.
4. In PSYC 110 Lecture 20, Professor Bloom describes depressed people randomly put into therapy or on a waiting list. Name this control condition and say what it can and cannot rule out.

Writing task

Research project: the variables and conditions of your experiment (Lessons 13-16 build your design and its validity).

Write one paragraph (150-200 words) for the method of your proposal on sleep and memory in university students, starting from your Lesson 6 outline. The running example: after learning 40 word

pairs, participants have a nap opportunity of up to 90 minutes or rest quietly while awake for the same time, then recall the pairs.

Include: your hypothesis (Lesson 5); the IV, how you will manipulate it and a short name for each condition; the DV and its measure (Lessons 10-12); three extraneous variables and how you will hold each constant; one possible confound and how random assignment or control prevents it. Use the future passive ("Participants **will be** randomly **assigned** ...").

If you work on your own topic, write the same parts for your question; if your IV cannot be manipulated, explain why (Lesson 17 offers a correlational alternative).

Self-editing checklist:

- One IV, with its levels called conditions.
- The IV is actively manipulated.
- The DV is measurable and clearly measures memory.
- Each extraneous variable has a specific method of control.
- One confound is named and dealt with.
- Every passive verb has *be* (will be assigned).
- Terms are defined; "research" has no plural.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PSYC 110 Lecture 15, Morality (diffusion of responsibility and the Kitty Genovese case): https://oyc.yale.edu/psychology/psyc-110/lecture-15
- PSYC 110 Lecture 20, The Good Life: Happiness (does therapy work? control groups and placebos): https://oyc.yale.edu/psychology/psyc-110/lecture-20

Answer key

1. Reading questions

1. The more witnesses there are to an accident or a crime, the less likely any of them is to help; the process is diffusion of responsibility (5.1, paragraphs 1-2).
2. IV: the number of other students participants believed were present; conditions: one, two or five other students; DV: whether the participant left the room to seek help (5.1, paragraph 4; 5.2, "What Is an Experiment?").
3. They were only consistent with the idea; there was no way to know whether the witnesses would have helped had there been fewer of them (5.1, paragraph 2).
4. Journal keeping was not manipulated. The groups already differed and probably differ in other ways (conscientiousness, introversion, stress), which are alternative explanations (5.2, "Manipulation of the Independent Variable").
5. Both data sets: happy $M = 4$, sad $M = 3$. Three sad-mood participants (scores 5, 7 and 5) recalled more than 4 (5.2, noise table).
6. It lowers external validity: results may not generalise beyond the people studied (5.2, paragraph after the table).
7. When average IQ differs across conditions; then IQ, not the IV, could explain the difference in the DV (5.2, "Extraneous Variables as Confounding Variables").
8. Placebo effects: the treatment group may improve because they expect to, and the no-treatment group does not (5.2, "Treatment and Control Conditions").

2. Book exercises

1. (example answer; many answers are correct) Can be manipulated: the number of people present, mood (with a happy or sad video clip), the writing topic, the treatment received (drug, placebo, therapy), the instructions given. Cannot be manipulated: age, gender, IQ, handedness, early illness experiences.
2. a. No: damaging people's brains is unethical, so researchers must study people who already have parietal damage (nonexperimental). b. No: depression cannot ethically be caused; researchers can only compare depressed and non-depressed people. c. Yes: teenagers can be randomly assigned to group training or a control condition, such as a wait-list. d. Yes: participants can be randomly assigned to a paid or an unpaid condition.
3. (example answer) The improvement may be a placebo effect: people in therapy expect to improve and people without treatment do not. Corrections: add a placebo control condition (for example, unstructured talk with a psychotherapist); use a wait-list control condition; or compare the therapy with the best available alternative, such as standard exposure therapy. Participants must be randomly assigned to conditions.

3. Language exercise

1. Each participant was isolated in a small room.

2. Some participants were told that the discussion involved only one other student.
3. The comments had been prerecorded and were played back.
4. The conditions are often given short descriptive names.
5. (example answer) A confounding variable is an extraneous variable that differs on average across the levels of the independent variable.

My notes